```
ASSIGNMENT 3


import csv



def read_attendance():
    records = []

    try:
        with open("attendance.csv", "r", newline="") as file:
            reader = csv.reader(file)

            for row in reader:
                if len(row) != 3:
                    print("Skipping malformed row:", row)
                    continue

                name, date, status = row

                name = name.strip()
                date = date.strip()
                status = status.strip().capitalize()

                if not name or not date:
                    print("Skipping invalid row:", row)
                    continue

                if status not in ["Present", "Absent"]:
                    print("Invalid attendance status:", row)
                    continue

                records.append([name, date, status])

    except FileNotFoundError:
        print("Error: attendance.csv not found.")
        return None

    return records
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def calculate_attendance(records):
    attendance = {}

    for name, date, status in records:
        if name not in attendance:
            attendance[name] = {
                "Present": 0,
                "Absent": 0
            }

        attendance[name][status] += 1

    return attendance



def calculate_percentage(attendance):
    percentages = {}

    for name, details in attendance.items():
        present = details["Present"]
        absent = details["Absent"]
        total = present + absent

        if total > 0:
            percentage = (present / total) * 100
        else:
            percentage = 0

        percentages[name] = percentage

    return percentages



def generate_report(attendance, percentages):
    try:
        with open("attendance_report.txt", "w") as file:
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                file.write("EMPLOYEE ATTENDANCE REPORT\n")
                file.write("=" * 35 + "\n\n")

                for name, details in attendance.items():
                    file.write("Employee: " + name + "\n")
                    file.write(
                        "Present days: " +
                        str(details["Present"]) + "\n"
                    )
                    file.write(
                        "Absent days: " +
                        str(details["Absent"]) + "\n"
                    )
                    file.write(
                        "Attendance percentage: " +
                        format(percentages[name], ".2f") + "%\n"
                    )

                    if percentages[name] < 75:
                        file.write(
                            "WARNING: Attendance below 75%\n"
                        )

                    file.write("\n")

            print("Attendance report generated successfully.")

        except OSError as e:
            print("Error writing report:", e)



    def add_attendance():
        name = input("Enter employee name: ").strip()
        date = input("Enter date (YYYY-MM-DD): ").strip()
        status = input("Enter status (Present/Absent):
    ").strip().capitalize()

        if not name or not date:
            print("Name and date cannot be empty.")
            return
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    if status not in ["Present", "Absent"]:
        print("Invalid status. Enter Present or Absent.")
        return

    try:
        with open("attendance.csv", "a", newline="") as file:
            writer = csv.writer(file)
            writer.writerow([name, date, status])

        print("Attendance record added successfully.")

    except OSError as e:
        print("Error adding attendance:", e)



def main():
    records = read_attendance()

    if records is None:
        return

    attendance = calculate_attendance(records)
    percentages = calculate_percentage(attendance)

    generate_report(attendance, percentages)

    choice = input(
        "Do you want to add a new attendance record? (yes/no): "
    ).strip().lower()

    if choice == "yes":
        add_attendance()


        records = read_attendance()

        if records is not None:
            attendance = calculate_attendance(records)
            percentages = calculate_percentage(attendance)
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                generate_report(attendance, percentages)


if __name__ == "__main__":
    main()



---------------------------------------------------------------------
attendence.csv

Anu,2026-10-01,Present
Anu,2026-10-02,Absent
Anu,2026-10-03,Present
Anu,2026-10-04,Present
Rahul,2026-10-01,Present
Rahul,2026-10-02,Absent
Rahul,2026-10-03,Absent
Rahul,2026-10-04,Absent
```